MIL-F-7194C(AV)
29 July 1981
Superseding
MIL-F-7194B
10 December 1974

MILITARY SPECIFICATION

FILTER, RECIPROCATING ENGINE INDUCTION AIR, AIRCRAFT

This specification is approved for use by US Army Aviation Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wet and dry engine induction air filters for air induction systems of aircraft reciprocating engines.

1.2 Classification. Filter shall be of the following types, as specified (see 6.2):

Type I Nonramming intake (subjected to low dust concentrations).

Type II Ramming intake (subjected to high dust concentrations).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for approval, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

P-D-680	Dry-Cleaning Solvent
PPP-B-601	Boxes, Wood, Cleated, Plywood
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Aviation Research and Development Command, ATTN: DRDAV-ERS, 4300 Goodfellow Boulevard, St. Louis, MO 63120, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2945

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MIL-P-116	Preservation-Packaging, Methods of
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-I-8599	Installation of Reciprocating Engine Induction Air Filters, Design Requirements for
MIL-L-21260	Lubricating Oil, Internal Combustion Engine, Preservation and Break-in
DOD-D-1000	Drawings, Engineering and Associated List

STANDARDS

FEDERAL

FED-STD-151	Metals: Test Methods

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Design. Engine induction air filter shall be designed for the air induction system of the designated aircraft in which the filter will be installed (see 6.2), and shall be in accordance with the type installations (see 1.2) as defined in MIL-I-8599. The proposed design for the filter shall have the approval of the procuring activity prior to start of production.

3.1.1 Preproduction sample. As soon as practicable after award of contract or order, a preproduction sample of the filter shall be tested to determine conformance with this specification (see 4.5). The preproduction sample shall be manufactured to the greatest extent possible by the use of tools, fixtures, and processes which will be used in the manufacture of filters on contract or order. This sample which has passed the tests shall be retained in the manufacturer's plant until all the filters on contract or order have been accepted by the Government. Approval of the preproduction sample by the contracting officer shall not relieve the contractor of his responsibility to supply filters conforming to this specification. Any changes or deviations from the preproduction sample shall be subject to the approval of the contracting officer.

3.2 Materials. Materials and processes used shall be of high quality, suitable for the purpose, and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used, provided the specifications are approved by the procuring activity and contain provision for adequate tests. The use of contractor's specifications shall not constitute waiver of Government inspection.

3.2.1 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service. The use of any protective coating that will crack or scale with age or extremes of climatic and environmental conditions shall be avoided. Metals shall be inspected, in accordance with FED-STD-151.

3.2.1.1 Dissimilar Metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.2.2 Recycled, Virgin and Reclaimed Materials. This specification shall require the use of reclaimed materials to the maximum extent possible (see 6.6).

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3.3 Construction. The basic filter unit construction and material shall be such that the unit will not cause combustion, and that it will withstand, without damage, the normal strains of jars, vibrations, backfire pressure, and other conditions incident to shipping, storage, installation, and service. Means shall be provided to prevent filter particles from entering the induction system.

3.3.1 Dimensions. Filter dimensions shall be governed by the air induction system design for the aircraft designated (see 6.2).

3.3.2 Standards and Specifications. Standards and specifications for commodities and services not specified herein, shall be selected in accordance with MIL-STD-143.

3.3.2.1 Standard Parts. AN and MS standard parts shall be used where they suit the purpose. Commercial parts, such as fasteners, connectors and cotter keys may be used, provided they can be replaced by AN and MS standards parts without alteration or degradation of function. The parts list shall reference the applicable AN and MS standard parts as acceptable replacements for the commercial parts.

3.3.3 Seal. If the specified aircraft installation requires a seal around the filter element, the seal and its method of attachment shall be of design and materials that will not be affected by normal servicing procedures. If the specified aircraft installation requires removal of filter element, means shall be provided for the purpose.

3.3.4 Drainage. Drain holes for draining an oil-wetted filter shall be located in a manner not to reduce the effectiveness of the filter.

3.3.5 Threaded parts. Threaded parts shall be in accordance with MIL-S-7742.

3.3.6 Positive locking. Internal and external threaded parts shall be positively locked or safetied.

3.3.7 Interchangeability of parts. Components shall be interchangeable (without requiring modification for replacement) with similar components furnished under the same contract or order.

3.4 Servicing instructions. Instructions for cleaning the filter shall be legibly and permanently marked in a conspicuous place on the filter frame.

3.4.1 Oil-wetted filter. Oil-wetted filter instructions shall read:

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"SERVICING INSTRUCTIONS"

"TO CLEAN: Wash thoroughly, soiled face down, in a cleaning fluid, and allow to dry. Dip in engine preservative oil conforming to grade 1 or 2 of MIL-L-21260, and allow to drain 2 to 4 hours. Wipe off excess oil.

3.4.2 Direction of airflow. Airflow shall be indicated by "AIR FLOW" and an arrow pointing in that direction.

3.5 Identification of product. Engine induction air filter shall be permanently and legibly stamped or marked in accordance with MIL-STD-130.

3.6 Drawings. If specified (see 6.2), one complete set of reproducible "as-constructed" drawings conforming to Class 2 of DOD-D-1000 shall be furnished the procuring activity concurrent with delivery of the filter. Lists of material shall be included on the drawings.

3.7 Workmanship. Engine induction air filter components shall fit properly in their respective positions. There shall be no cracks, breaks, chips, bends, burrs, loose attaching parts, or any other defects that could adversely affect the life, use, or suitability of the filter. All surfaces shall be smooth, clean, and free from corrosion, dirt, and foreign material. All surfaces with protective coatings shall be free of any defects which could affect appearance or serviceability (see 4.4 and Table I).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be inspected in accordance with 4.4 and tested in accordance with 4.5. The manufacturer's certification that the filter has previously passed the tests specified herein may be accepted instead of the actual performance of the tests, if this certification is submitted, the report of test shall be in accordance with MIL-STD-831, and shall be submitted to the contracting officer prior to production.

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4.3 Sampling. All filters offered for delivery at one time shall be considered a lot for purposes of inspection. A random sample of filters shall be selected from each lot in accordance with level II of MIL-STD-105.

4.4 Examination. Defects shall be classified as indicated in Table I. Each filter selected in accordance with 4.3 shall be visually examined to verify compliance with this specification. Acceptable quality level (AQL) shall be 2.5 percent for major defects and 6.5 percent for minor defects.

TABLE I. Classification of defects in accordance with MIL-STD-105

Examine	Defect	Major	Minor
Finish.....	Blistered, peeled, chipped, or area of no film or thin film, affecting serviceability or appearance.....	X	
	Poor touch-up, orange peel, color separation, smudge, fish scale, stain, over-spray, sag, imprint or foreign matter embedded, or finish otherwise impaired, affecting appearance.....	...	X
	Any area of rust.....	...	X
Materials.....	Any component not fabricated of the material authorized in the approved sample.....	X	
Type.....	Not type specified in contract	X	
Design.....	Any characteristic not in accordance with the specified requirements.....	X	
Construction and workmanship (applicable to all components and assemblies).	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, decayed, or otherwise impaired material or workmanship, affecting serviceability.....	X	
	Defective or otherwise damaged, but not affecting serviceability.....	...	X
	Misplaced or loose parts or parts not in proper alinement, but not affecting serviceability.....	...	X

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	Any sharp burr, sliver, or splinter that may be injurious to personnel or may affect serviceability.....	...	X
	Any operation omitted or not properly performed, affecting serviceability.....	X	
	Any operation omitted or not properly performed, but not affecting serviceability.....	...	X
	Not connected or joined as specified, or poorly accomplished, affecting serviceability.....	X	
Assembly.....	Any component not properly assembled or secured, affecting serviceability, such as media damage, media not properly installed in frame creating passage of air without filtering.....	X	
	Unit perceptibly out of square of alinement.....	X	
Identification...	Missing, incomplete, not legible, not specified type or size, not affixed in the prescribed manner.....	...	X

4.5 Tests. Filter tests shall be for (1) initial efficiency as specified in 4.5.1; (2) dust holding capacity as specified in 4.5.1.5; and pressure drop as specified in 4.5.1.7.

4.5.1 Preproduction. The preproduction sample shall be prepared as specified in 4.5.1.2 and 4.5.1.3. Prior to preproduction tests, sufficient tests shall have been conducted to determine the highest velocity of airflow at which the prescribed values of efficiency and pressure drop (see Table II) can be met. This velocity, expressed in feet per minute (fpm), shall determine the rating of the filter. No filter with a rating of less than 1,000 fpm will be considered for approval. At the option of the procuring activity, preproduction tests may be supplemented with tests under actual service conditions.

4.5.1.1 Procedures.

4.5.1.2 Dry filter. Prior to test, filter shall be cleaned with solvent conforming to Specification P-D-680, in accordance with instructions stamped on the filter frame (see 3.4).

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4.5.1.3 Oil-wetted filter. Oil-wetted filter shall be thoroughly cleaned of oil and grease or other corrosion preventive compounds (see 4.5.1.2). The filter shall be thoroughly dried, then dipped in engine preservative oil conforming to grade 1 or 2 of Specification MIL-L-21260 followed by a 20-hour draining period (± 15 minutes) at 21.1° to 32.2°C (70° to 90°F). Filter shall then be installed in the test setup (see 4.5.1.6), and clean air at 21.1° to 32.2°C (70° to 90°F) forced through the filter at the rated velocity of the filter, for 30 minutes, to remove excess oil.

4.5.1.4 Dusts. Test dusts shall be in accordance with Table II.

TABLE II. Pressure drop and efficiency test (dust particle size)

Test No.	Dust Description	Particle size (microns)	Percent of that size
1	100 percent prepared dust, AC Spark Plug Co. Part No. 1543637, or equal	0-5	12, ± 2
		5-10	12, ± 3
		10-20	14, ± 3
		20-40	23, ± 3
		40-80	30, ± 3
		80-200	9, ± 3
2	70 percent of dust from test No. 1, and 30 percent prepared Arizona road dust, AC Spark Plug Co. Part No. 1543094, or equal. Table shows Arizona road dust composition.	0-5	39, ± 2
		5-10	18, ± 3
		10-20	16, ± 3
		20-40	18, ± 3
		40-80	9, ± 3
3	50 percent of dust from test No. 1 and 50 percent of Arizona road dust from test No. 2.		
4	100 percent high silica beach sand, graded by dry screening, with 20-minute shaking period.	(U.S. std sieve)	
		-50 + 70 mesh	45, ± 3
		-70 + 100 mesh	20, ± 3
		-100 + 140 mesh	15, ± 3
		-140 + 200 mesh	10, ± 3
		-200 + 325 mesh	10, ± 3

Note. Dust concentration for type I filter shall be 0.0075-gram per cubic foot; for type II filters, 0.03-gram per cubic foot.

4.5.1.5 Particle size. A dust designated as -50 + 70 (see Table II) shall consist of particles of all which pass through a #50-mesh screen and all of which are retained on a #70-mesh screen.

4.5.1.6 Test setup. A test filter setup having an effective frontal area of 6 by 6 inches, as shown on Figure 1, shall be used for the pressure drop and efficiency test specified in 4.5.1.7. This test setup, except for the size of the effective filter face, shall be identical in construction to the filter to be tested.

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4.5.1.7 Pressure drop. The filter shall be tested using the different types of dust specified in Table II. Following each test, the filter shall be prepared for the succeeding test as specified in 4.5.1.2 and 4.5.1.3, as applicabl Each test shall consist of 12 runs of 10-minute duration. During the first, fifth, and ninth runs, dust shall be fed to the airstream by the aspiration method to reduce agglomeration of particles and at a uniform rate in concentrations as shown in Table II. Type I filter shall be subjected to tests Nos. 1, 2, and 3, and type II filter tests Nos. 1 and 4, of Table II.

4.5.1.7.1 Following each 10-minute run, the cleaning efficiency (see 4.5.1) and pressure drop across the filter shall be recorded. Efficiencies determined immediately after a 10-minute period of dust introduction shall be based on the amount of dust stopped by an absolute filter which shall be installed on the downstream side of the filter being tested, and the amount of dust fed into the airstream during that particular 10-minute period. Efficiencies determined immediately after a 10-minute clear air run shall be based on the amount of dust fed to the airstream during the preceding 10-minute period of dust introduction. The total amount of dust stopped by the absolute filter will be indicated by the weights obtained after the clean air run in question.

4.5.1.7.2 Upon completion of this test, and at any other time during the tests the pressure drop through the filter shall not be more than 8 inches of water, and the minimum filter efficiency shall be 90 percent.

4.6 Preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to assure compliance with Section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Filters shall be preserved and packaged level A or commercial, as specified (see 6.2).

5.1.1 Level A. Filters shall be cleaned in accordance with Method C-1 of MIL-P-116. Oil-wetted filters shall then be dipped in MIL-L-21260, grade 1 or engine preservative oil. Excess oil shall be drained for a 20-hour period or dispelled by any suitable process. Preservative is not required on dry filters Each filter shall be packaged Method 1A-13 of MIL-P-116.

5.1.2 Commercial. Filters shall be preserved and packaged in accordance with MIL-STD-1188.

5.2 Packing. Filters preserved and packaged in accordance with 5.1 shall be packed level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. Filters shall be grouped and packed in snug-fitting plywood boxes conforming to PPP-B-601. Intermediate containers shall be utilized as necessary.

5.2.2 Level B. Filters shall be packed as for level A except that PPP-B-636, class weather-resistant, fiberboard boxes may be used.

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5.2.3 Commercial. Filters shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special markings required by the contract or delivery order, marking shall be in accordance with MIL-STD-129 or MIL-STD-1188, as specified (see 6.2).

6. NOTES

6.1 Intended use. Filter is intended to provide filtered air for reciprocating engine-powered aircraft air induction systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Specified aircraft and type of installation for which filter is required (see 1.2 and 3.1).

(c) Whether drawings are required (see 3.6).

(d) Instructions for delivery of data (see 6.5).

(e) Level of packaging and preservation required (see 5.1).

(f) Level of packing required (see 5.2).

(g) Marking(s) required (see 5.3).

6.3 DEFINITIONS

6.3.1 Nonramming intake filter. A nonramming intake filter is a filter installed in a duct system, of such a design and location that a major portion of the dust in the air bypasses the duct entrance and is not free to impinge on the face of the filter.

6.3.2 Ramming intake filter. A ramming intake filter is a filter installed in a duct system, of such a design and location that the major content of the dust in the air is free to impinge on the face of the filter.

6.4 Certification. Certification requires the supplier to furnish a document properly executed by the supplier, stating that all requirements of this specification, including its referenced documents, and the contract itself, have been met.

6.5 Data. Data generated by this document shall not be delivered to the procuring activity unless specified on the Contract Data Requirements List (DD 1423).

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6.6 Reclaimed Materials. This specification shall require the use of reclaimed materials to the maximum extent possible without jeopardizing the intended end use of the item (see 3.2.2).

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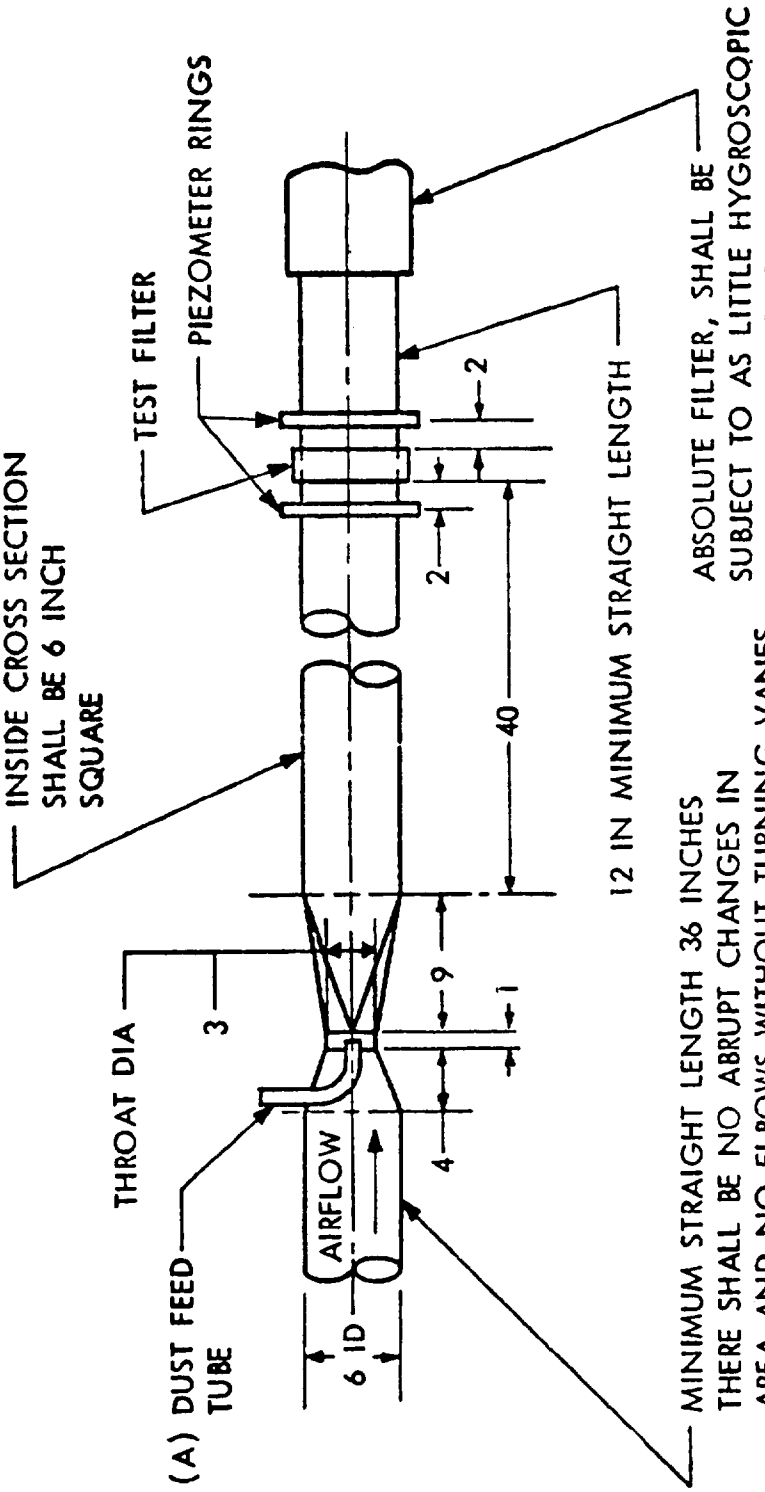


FIGURE 1. *Air filter test duct.*

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